

Question One

(a) Solve $T(n) = 4T(n/6) + n$, by master method

$$a = 4, b = 6, f(n) = n$$

$$\exists \epsilon = (1 - \log_6 4) > 0 \text{ such that}$$

$$f(n) = n \geq n^{\log_6 4 + (1 - \log_6 4)} = n^{\log_6 a + \epsilon}$$

$$f(n) = \Omega(n^{\log_6 a + \epsilon})$$

Regularity check

$$\exists c = 4/6 < 1$$

$$af(n/b) = 4(n/6) = (4/6)n = cn = cf(n)$$

case 3 applies

$$T(n) = \Theta(f(n))$$

$$T(n) = \Theta(n)$$

$$T(n) = O(n)$$

b) Solve $T(n) = 4 T(n/6) + n$, by expansion.

$$\begin{aligned} T(n) &= 4T(n/6) + n \\ &= 4[4T(n/6^2) + n/6] + n \\ &= 4^2 T(n/6^2) + n[1 + (4/6)] \\ &= 4^2 [4T(n/6^3) + (n/6^2)] + n[1 + (4/6)] \\ &= 4^3 T(n/6^3) + n[1 + (4/6) + (4/6)^2] \\ &= 4^k T(n/6^k) + n[1 + (4/6) + (4/6)^2 + \dots + (4/6)^{k-1}] \\ &= n^{\log_6 4} T(1) + n[1 + (4/6) + (4/6)^2 + \dots + (4/6)^{k-1}] \\ &< n^{\log_6 4} + n[1 + (4/6) + (4/6)^2 + \dots] \\ &= n^{\log_6 4} + n[1 / (1 - (4/6))] \\ &= O(n) \end{aligned}$$

$$\text{Setting } n = 6^k.$$

$$\text{Since } 4 = 6^{(\log_6 4)}, \text{ where the base of log is 6,}$$

$$4^k = [6^{(\log_6 4)}]^k = (6^k)^{\log_6 4} = n^{\log_6 4}$$

c) Solve $T(n) = 4 T(n/6) + n$, by MI.

Claim: $T(n) < cn, c > 3, n > 6$.

$$T(n) = 4 T(n/6) + n$$

$$< 4 c(n/6) + n$$

$$= (2/3)cn + n$$

$$< cn, \text{ since } c > 3$$

d) Solve $T(n) = 8 T(n/6) + n$, by master method

$$a = 8, b = 6, f(n) = n$$

$$\exists \epsilon = (\log_6 8 - 1) > 0 \text{ such that}$$

$$f(n) = n^1 = n^{\log_6 8 - (\log_6 8 - 1)} = n^{\log_6 a - \epsilon}$$

$$f(n) = O(n^{\log_6 a - \epsilon})$$

Case 1 applies

$$T(n) = \Theta(n^{\log_6 a})$$

$$T(n) = \Theta(n^{\log_6 8})$$

$$T(n) = O(n^{\log_6 8})$$

e) Solve $T(n) = 8 T(n/6) + n$, by expansion.

$$T(n)$$

$$= 8T(n/6) + n$$

$$\begin{aligned} &= 8[8T(n/6^2) + n/6] + n \\ &= 8^2 T(n/6^2) + n[1 + (8/6)] \\ &= 8^2 [8T(n/6^3) + (n/6^2)] + n[1 + (8/6)] \\ &= 8^3 T(n/6^3) + n[1 + (8/6) + (8/6)^2] \\ &= 8^k T(n/6^k) + n[1 + (8/6) + (8/6)^2 + \dots + (8/6)^{k-1}] \\ &= n^{\log_6 8} T(1) + n[((8/6)^k - 1) / ((8/6) - 1)], \text{ base 6 log} \\ &= n^{\log_6 8} + [(n^{\log_6 8} - n) / ((8/6) - 1)] \\ &= O(n^{\log_6 8}) \end{aligned}$$

$$\text{Setting } n = 6^k.$$

$$\text{Since } 8 = 6^{(\log_6 8)}, \text{ where the base of log is 6,}$$

$$8^k = [6^{(\log_6 8)}]^k = (6^k)^{\log_6 8} = n^{\log_6 8}$$

$$(8/6)^k = (8^k)/(6^k) = (n^{\log_6 8})/n$$

f) Solve $T(n) = 8 T(n/6) + n$, by MI

Claim $T(n) < c n^{\log_6 8} - c'n$,

$$c > c', c' > 3, n > 6$$

$$T(n)$$

$$= 8 T(n/6) + n$$

$$< 8[c(n/6)^{\log_6 8} - c'(n/6)] + n$$

$$= 8c[n^{\log_6 8}]/[6^{\log_6 8}] - 8c'(n/6) + n$$

$$= 8c[n^{\log_6 8}]/8 - c'n - (2/6)c'n + n$$

$$= cn^{\log_6 8} - c'n - (c'/3 - 1)n$$

$$< cn^{\log_6 8} - c'n, \text{ since } c' > 3.$$

Question Two:

$$f(n) = n / \log n, a \geq 1, \text{ and } b > 1.$$

a) Prove that $\forall k \geq 0, f(n) \neq \Theta(n^{\log_b a} \log^k n)$.

Assume for contradiction that

$$\exists k \geq 0, f(n) = \Theta(n^{\log_b a} \log^k n).$$

$$\text{Then, } n / \log n = \Omega(n^{\log_b a} \log^k n)$$

For large n , and for some c ,

$$n / \log n > c n^{\log_b a} \log^k n$$

$$n > c n^{\log_b a} \log^{k+1} n$$

$$n^{1 - \log_b a} > \log^{k+1} n$$

Since $k+1 > 0$, we have $a < b$.

Then, for any $c > 0$, for large n ,

$$c n^{\log_b a} \log^{k+1} n < n,$$

$$c n^{\log_b a} \log^k n < n / \log n,$$

$$\text{Hence, } n / \log n \neq O(n^{\log_b a} \log^k n)$$

$$\text{Hence, } f(n) \neq \Theta(n^{\log_b a} \log^k n)$$

b) It is NOT feasible to use master method for solving

$$T(n) = 2T(n/2) + n / \log n.$$

From (a), Case 2 cannot apply.

Since $f(n) = n / \log n$, and $\log_b a = 1$,

For all $\epsilon > 0, f(n) = n / \log n > n^{\log_b a - \epsilon}$,

and $f(n) = n / \log n < n^{\log_b a + \epsilon}$.

Hence, Cases 1 and 3 cannot apply.